Sahaj Bhadja

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Summary

Platform Engineer with 2+ years building **Python/C++ services**, **robot data pipelines**, telemetry, observability, cloud integrations, deployment tooling, and autonomy software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **fault-tolerant asynchronous services** with FastAPI workers, explicit job states, timeouts, failure isolation, and monitoring, processing 1K+ concurrent jobs at 3x throughput without cascading system failures under sustained load.
- Built data-upload pipelines across Azure SQL, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with **validation, reconciliation, CI/CD, and 99%+ production reliability** across field-facing environments.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed **concurrent Python data-processing services** for secure PDF, DOCX, and ZIP ingestion, applying ThreadPoolExecutor, OCR, timeouts, and model fallbacks while processing 1K+ files per batch under peak load.
- Built React and Flask monitoring workflows for uploads, processing history, failures, and feedback, supporting 10K+ daily requests through **REST APIs, automated testing, CI/CD, and production debugging** for daily operations.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for **autonomous-driving control** in PyTorch, processing 100K+ trajectory logs through reproducible evaluation pipelines and improving long-horizon planning performance 18% in simulation for autonomy research.
- Integrated spatial-temporal encoding across HD-map geometry, lane context, and ego trajectories, using multi-head attention to improve **closed-loop control stability** 15% across simulated driving baselines during reproducible autonomy experiments.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Debugged distributed failures across Windows, Linux, and macOS while engineering microservices and APIs, improving response times 30% through **reproducible builds, structured logs, automated validation, and cross-platform testing**.
- Automated **deployment and field-test workflows** with 20+ PyTest and Mockito suites, Jenkins pipelines, VM provisioning, and cross-platform execution, increasing pipeline efficiency 68% and reducing technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight microservices for authentication, resources, billing, and notifications, using **low-latency WebSocket communication** to synchronize 2M+ records and real-time state changes across distributed locations for operational workflows.
- Built operational dashboards and supported eight production services through **Docker images, Jenkins CI/CD, Sentry monitoring, structured logging, incident triage, and reliable troubleshooting** across distributed customer-facing production environments.

Skills

Languages: Python, C++, TypeScript, SQL
Robotics & Communication: ROS2, gRPC, MQTT, DDS, WebSockets, IPC, Protocol Buffers, Sensor Data, Control Integration
Data & Cloud: Amazon Web Services, GCP, S3, GCS, PostgreSQL, SQL Server, Streaming Pipelines, Cloud Storage
Infrastructure: Docker, Kubernetes, Linux, CI/CD, Jenkins, GitHub Actions, Git, Distributed Systems, OTA Updates
Reliability: PyTest, Unit Tests, Integration Tests, Debugging, Monitoring, Logging, Telemetry, Observability, Fault Isolation, Validation

Projects

Robot Network | *C++, Python, ROS2, gRPC, MQTT, Protocol Buffers, Docker*

- Designed a **robot communication backbone** connecting sensor, control, teleoperation, and cloud services through typed messages, health checks, bounded queues, reconnection policies, timestamp validation, and fault-isolated ROS2 components.

Data Relay | *Python, C++, WebSockets, S3, GCS, Kubernetes, Monitoring*

- Built a **robot telemetry and data pipeline** that buffers sensor streams, validates schemas, uploads partitioned datasets, tracks fleet health, exposes real-time diagnostics, and supports staged over-the-air software rollouts.

CPU-scheduling Simulator | *C++, React, Scheduling Algorithms, Performance Analysis* | Live

- Implemented **real-time scheduling simulations** for FCFS, SJF, SRT, Round Robin, priority, and multilevel queues with dynamic controls, execution visualization, and correctness testing across 1K+ process combinations under varied workloads.